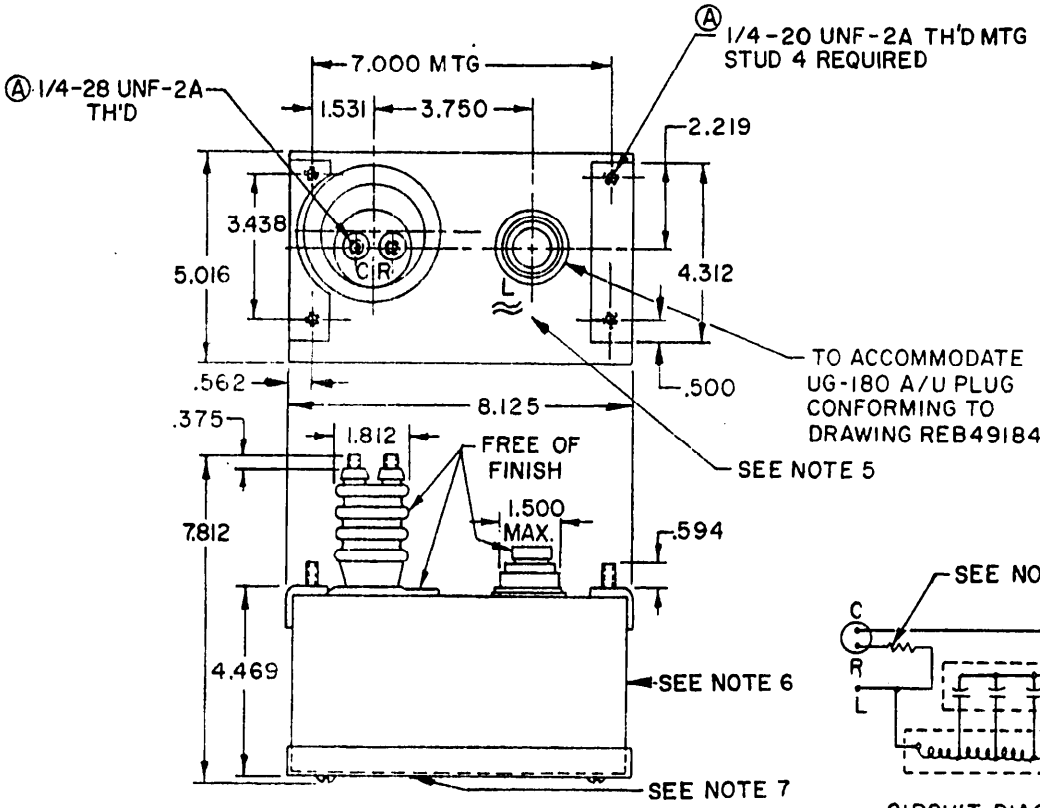
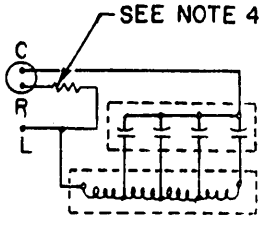
MILITARY SPECIFICATION SHEET

# NETWORK, PULSE FORMING, RADAR, TYPE PN4E331J142Q350

The complete requirements for procuring the pulse forming network described herein shall consist of this document and the latest issue of Specification MIL-N-23182.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.




CIRCUIT DIAGRAM

| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .375 | 9.53 | 3.438 | 87.33 |
| .500 | 12.70 | 3.750 | 95.25 |
| .562 | 14.27 | 4.312 | 109.52 |
| .594 | 15.09 | 4.469 | 113.51 |
| 1.500 | 38.10 | 5.016 | 127.41 |
| 1.531 | 38.89 | 7.000 | 177.80 |
| 1.812 | 46.02 | 7.812 | 198.42 |
| 2.219 | 56.36 | 8.125 | 206.38 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.015 (.38 mm).
4. A device intended to discharge the capacitors for personnel protection.
5. Stamp in .125 (3.18 mm) letters 'Caution: Use only as output terminal.'
6. Stamp in .125 (3.18 mm) letters 'Operate with this side up'.
7. Stamp in .125 (3.18 mm) letters 'Manufacturer's name and Government designation.

Case and mounting dimensions, and circuit diagram.

Ⓐ denotes changes



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REQUIREMENTS:

Dimensions and configuration: See figure.

Case: Metal.

Circuit diagram: See figure.

Maximum operating temperature: +85° C.

Pulse repetition frequency:
1, 350 pulses per second.

Total capacitance: 0.004 ± 0.0004 microfarad (uf).

Load impedance: 35 ohms.

Supply voltage (to pulse forming network):
8, 000 vdc.

Charging voltage: 16, 000 vdc.

Terminals: Screw.

Terminal strength:
Method 211, test condition E, of MIL-STD-202.
The following detail shall apply:
Torque: 32 pound-inches.

Sealing: As specified in MIL-N-23182.
Networks shall be immersed for 2 to 3 minutes in a bath of water or any other liquid of no greater viscosity maintained at a temperature of at least +85° C. The temperature of the network shall not exceed +40° C at the time of immersion. During the immersion, networks shall show no flow of air or gas bubbles (other than an occasional entrapped air or gas bubble), or leakage of compound.

Corona discharge:
As specified in MIL-N-23182, at atmospheric pressures of from 29.92 inches of mercury (sea level) to 20.58 inches of mercury (10, 000 feet) with no indication of corona.

Dielectric withstanding voltage:
As specified in MIL-N-23182.
Pulse-forming networks shall withstand the specified potentials without breakdown, puncturing of insulation or arcing.

NOTE: After completing the dielectric withstanding voltage test the discharge circuit shall remain connected to the network (terminals C and R strapped) for a sufficient time (approximately 10 minutes) to insure that the voltage across terminals C and L with the protection removed shall not exceed 25 volts.

Barometric pressure (reduced):
Method 105 of MIL-STD-202. The following details and exceptions shall apply:
Atmospheric pressure: 20.58 inches of mercury (10, 000 feet).
Application of test voltage: 125 percent of rated charging voltage for at least 1 minute between each terminal and every other terminal in turn, and between the container and each terminal not connected to the container.
After the test, networks shall be visually examined for evidence of short circuiting, arcing, breakdown of insulation and other visible damage.

Insulation resistance:
Method 302, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Electrification time: 1 minute, maximum.
Terminal to terminal (terminal L to terminal C).
At 25° C: 500 megohm-microfarads, minimum.
At +85° $^{+3°}_{-0°}$ C: 15 megohm-microfarads, minimum.
Terminals to container (where container is not a terminal), at 25° C: Shall exceed 3, 000 megohms.

Dissipation factor at 25° C:
As specified in MIL-N-23182. The following details and exceptions shall apply:
Frequency of measurement (or reference): 1, 000 Hz.
Test potential: 500 volts root mean square (rms), maximum.
Dissipation factor: Shall not exceed 1 percent.

Capacitance (terminal to terminal):
Method 305 of MIL-STD-202. The following details and exceptions shall apply:
Test frequency: 1, 000 Hz ± 100 Hz.
Test potential: 500 volts rms, maximum.
Limit of accuracy of measurement: ±2 percent.
Total measured capacitance: Within the limits of 0.0043 to 0.0051 uf.



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Waveshape electrical characteristics:
Measured as specified in MIL-N-23182, and shall be as follows:
Pulse duration: 0.33 microseconds (μsec) ±0.03 μsec.
Percent ripple: 15 percent, maximum.
Pulse rise time: 0.025 μsec maximum.
Pulse decay time: 0.075 μsec, maximum.
Characteristic impedance: 35 ohms.
Duty cycle: Between 0.000405 and 0.000486.

Early life survival:
As specified in MIL-N-23182. The following details and exceptions shall apply:
Prior to the test, networks shall be subjected to requirements, at standard test conditions, for corona discharge and capacitance.
Terminating resistance: Not to exceed 26.25 ohms.
Test temperature: Not less than +85° C.
Pulse repetition frequency: 338 pulses per second.
At the conclusion of the 24-hour exposure, networks shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge. In addition, capacitance shall be measured at 25° C. Capacitance change, from the initial measurement, shall not exceed ±2 percent.

Thermal shock:
Method 107, test condition A, of MIL-STD-202. The following details and exceptions shall apply:
Exposure time:
At $-55^{\circ}\ ^{+0^{\circ}}_{-3^{\circ}}$ C: 2 hours.
At $+85^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C: 2 hours.
After completion of the final cycle, networks shall be examined for evidence of leakage or other visible damage. In addition, networks shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, capacitance, and waveshape electrical characteristics.

Moisture resistance:
Method 106, figure 106-1, of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: On racks by normal mounting means.
Conditioning: Networks shall undergo a 24-hour initial drying period prior to first cycle.
Polarization voltage: 100 vdc, during steps 1 to 6 inclusive between all terminals not connected directly to each other.
Upon completion of step 6 of the final cycle, networks shall be removed from the humidity chamber and shall be examined for evidence of leakage, or other visible damage. Networks shall then be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, capacitance, and waveshape electrical characteristics.

Vibration:
Method 201 of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: By normal means.
Planes of vibration: As specified in MIL-N-23182 for units weighing more than 10 pounds.
Measurements during vibration: During the last one-half hour of vibration, a suitable measurement shall be made to detect intermittent contacts, or open- or short-circuiting.
After the test, networks shall be examined for evidence of leakage, or mechanical damage. Networks shall then be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, and capacitance.

Shock (medium impact):
Method 205, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: By normal means.
Number and direction of applied shocks: 30 impact shocks, 10 in each of three mutually perpendicular planes.
After the test, networks shall be examined for evidence of physical or mechanical damage. In addition, networks shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, and capacitance.



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Low temperature:
As specified in MIL-N-23182.
Networks shall be subjected to a temperature of -55° $^{+0°}_{-3°}$ C for 48 hours with rated charging voltage applied, with no evidence of breakdown.

Capacitance change at reduced temperature:
As specified in MIL-N-23182. The following details and exceptions shall apply:
At the conclusion of the low temperature test, and while networks are still subjected to the -55° C temperature, the capacitance of the pulse forming networks shall be measured. The measurement shall be made at, or referred to, a frequency of 1,000 Hz. The test potential shall be 500 volts rms or less. Capacitance change shall not exceed -5 percent.

Salt spray (corrosion):
Method 101, test condition B, of MIL-STD-202.
The following details and exceptions shall apply:
Salt solution concentration: 5 percent.
After the test, networks shall be examined for evidence of corrosion and obliteration of marking.

Life:
As specified in MIL-N-23182 for networks with special life requirements. The following details and exceptions shall apply:
Exposure time: 1,000 hours.
Exposure temperature: +85° $^{+3°}_{-0°}$ C.
Test potential: 16,000 vdc.
Input power: 3,258 volt-amperes.
Temperature rise, as specified in MIL-N-23182, shall not exceed 30° C.
After the test, networks shall be subjected to, and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, and waveshape electrical characteristics. In addition, networks shall be examined for evidence of physical or electrical damage.

Part number:
Consists of the basic number of this specification sheet and a dash number.

M23182/2-01

| Basic document number | Dash number |
|---|---|

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - MI, SM
Navy - MC, AS, OS
Air Force - 19

Preparing activity:
Navy - EC

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